Financial Management

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Module 3: Investment Decisions and Risk Analysis

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3.1:Cost of Capital

Topics Covered

- Understanding the relevance and application of cost of capital to business decisions
- Determine the cost of various components of Capital, including debt, and equity
- Calculate the Weighted Average Cost of Capital (WACC)
- Impact of floatation cost on cost of capital

What is Cost of Capital – Hurdle Rates

- The discount rate used for calculating the present value of future cashflow is also called as the **Cost of Capital.**
- ***The return that an investor expect from any investment/securities is the cost of capital for that company that issues.***
- The Financing Decision of Financial Management calls for sourcing of fund for the business which decides Capital Structure of the company.
- There are mainly two components of this Capital : Equity and Debt
- Equity Capital includes equity shares, preference shares and retained earnings or reserves and surplus.
- Debt Capital includes long-term and short-term borrowings and debt security.

Applications of Cost of Capital

- Evaluates investment projects
- Determines the capital structure
- Assesses leasing proposals
- Sets the rates that regulated organisations

Calculating Cost of Capital
1. Cost o Debt, k_d

1. **Cost of Debt (k_d):** cost of debt instrument is the yield to maturity of that instrument.

$$P_0 = \sum_{t=1}^{n} \frac{I}{(1+k_d)^t} + \frac{F}{(1+k_d)^n}$$

Where,

P_0 = current market price of the debentures

I = annual interest payment

n = number of years left to maturity

F = maturity value of the debenture

- computation of k_d requires a trial and error procedure

OR

- You can use following formula which gives a very close approximation to the correct value

$$k_d = \frac{I+(F-P_0)/n}{0.6P_0+0.4F}$$

Exercise

Abacus Ltd. Issued 15 years, 14 per cent bonds five years ago. The bond which has a face value of Rs. 100 is currently selling for Rs. 108.

a. What is the pre-tax cost of debt?
b. What is the after-tax cost of debt? (assume a 35 per cent tax rate)

Cost of Capital
2. Cost of Preference Shares, k_p

2. Cost of Preference Shares (k_p) Redeemable: Cost of preference share is equal to its yield.

$$k_p = \frac{D+(F-P_0)/n}{0.6P_0+0.4F}$$

Where,

P_0 = current market price of the preference share

D = annual dividend payment

n = number of years left to maturity

F = maturity value of the preference shares

If the Preference share is irredeemable then

$$k_p = D/P_0$$

Exercise

Omega Enterprises issued 10 year, 9 per cent preference shares four years ago. The preference share which has a face value of Rs.100 is currently selling for Rs. 92. what is the cost of preference shares?

Kp = [D+(FV-P0)/n]/[0.6P0+0.4FV]

Kp = [9 + (8/6)]/(55.2+40)] = 10.85%

If, 9 % PS irredeemable

Kp = D/P0 = 9/92 = 9.78%

Cost of Capital

3. Cost of Retained Earnings (Kre)/Equity, k_e

Cost of Retained Earnings/Equity (k_e): is the rate of return on equity that require on equity finance (external equity) that firm obtains by way of retained earnings (Internal equity)

i) The Dividend Growth Model Approach (Gordon Constant Growth Model): Price of an equity stock generally depends on the dividends expected on it.

$$k_e = \frac{D_1}{P_0} + g$$

These dividends are expected to grow at a constant rate of ***g*** per cent per year.

$$k_e = \frac{D_0(1+g)}{P_0} + g$$

Where,

$\frac{D_1}{P_0}$ = dividend yield

g = expected growth rate

Exercise

The market price per share of a company is Rs.17.00. The dividend expected a year hence is Rs.1.00. The expected rate of dividend growth is 8 per cent. What is the cost of equity capital of the company?

P0 = 17

D1=1

g = 8%

Ke = ?

Ke = (1/17) +0.08 = 13.88%

Suppose, dividend paid last year is 0.8, calculate Ke?

D0 = 0.8

D1 = D0*(1+g) = 0.8 * (1+0.08) = 0.864

Ke = (0.864/17) + 0.08 = 13.08%

Cost of Capital
3. Cost of Retained Earnings, k_e

ii) The Capital Asset Pricing Model (CAPM) Approach: calculates the expected return on a security based on its level of risk.

$$k_e = r_f + \beta (r_m - r_f)$$

Where,

r_f **= risk free rate :** this may be estimates as the yield on a long-term government bond that has a maturity of 10 years or more

β = coefficient of equity: beta is the measure of risk involved with investing in a particular stock relative to the risk of the market. This may be calculated by regressing the monthly returns on the stock over the monthly returns on the market index over the past 60 months or more.

r_m **= return on the market:** this may be estimated by a broad stock market index like the BSE National Index, NSE

$r_m - r_f$ **= market risk premium:** this may be estimated as the difference between average return on the market portfolio and the average risk free rate over the past 10 to 30 years – the longer the period, the better it is.

Exercise

The risk-free rate is 8 per cent and the return on market portfolio is 12 per cent. Calculate the required rate of return on the three equity stocks with the beta values shown against them.

Stock	Beta
A	0.8
B	1.2
C	1.7

Rf = 8%

Rm = 12%

Ke = A, B, C

Ke (A) = 8% + (0.8 * (12%-8%) = 11.2%

Ke (B) = 12.8%

Ke (C) = 14.8%

Cost of Capital
Cost of External Equity, k_e

Cost of External Equity (k_e) : The cost of external equity is higher than the cost of retained earnings because when a company issues external equity it incurs flotation or issue costs.

$$k_e = \frac{D_1}{P_0\,(1 - F)} + g$$

Where

k_e = cost of external equity

F = percentage flotation cost incurred in issuing new equity

Exercise

The current market price per share of Vishnu Ceramics is Rs.75. The dividend expected a year from now is Rs. 2.5 and it is expected to grow at a constant rate of 12 per cent. The floatation cost for the new issue will be 8 per cent. What is the cost of new equity?

P0 = 75

D1 = 2.5

g = 12%

F = 8%

Ke = ???

Ke = [D1/(P0*(1-F)] + g = 15.62%

Weighted Average Cost of Capital (WACC) = k0

- Given the cost of specific sources of financing and the scheme of weighting, the WACC can be readily calculated by multiplying the specific cost of each source of financing by its proportion in the capital structure and adding the weighted values.

$$\textbf{WACC} = w_e\ k_e + w_p\ k_p + w_d\ k_d\ (1 - t)$$

Where,

w_e = proportion of equity

w_s = proportion of retained earnings

w_p = proportion of preference

w_d = proportion of debt

t = corporate tax rate

Practice Problem

The capital structure of V K Funds Ltd. in book value terms is a follows

Source of Finance	Amount in Millions
Equity Capital (20 million shares, Rs.10/share)	Rs. 200 M
Preference Capital, 12 per cent (500,000 shares, Rs. 100/share)	Rs. 50 M
Retained earnings	Rs. 350 M
Debentures 14 per cent (1200000 debentures, Rs. 100 par)	Rs. 120 M
Term Loans, 13 per cent	Rs. 80 M
Total	**Rs. 800 M**

Practice Problem

The next expected dividend per share is Rs. 2. The dividend per share is expected to grow at the rate of 12 per cent. The market price per share is Rs. 50. Preference stock, redeemable after 10 years, is currently selling for Rs. 85 per share. Debentures, redeemable after 5 years, are selling for Rs. 90 per debenture. The tax rate for the company is 30 per cent. Calculate the weighted average cost of capital.

Mini Case

Sikha Jaishwal, Managing Director of FaSon Textiles, was reviewing two very different investment proposals. The first one is for expanding the capacity in the main line of business and the second one is for diversifying into a new line of business.

Sikha Jaishwal asks for your help in estimating FaSon's weighted average cost of capital which she believes is relevant for evaluating the expansion proposal.

The latest balance sheet of FaSon is given below

Liabilities + OE	Amount	Assets	Amount
Equity Capital	350	Fixed Assets	700
Preference Capital	100	Investments	100
Reserve and Surplus	200	Current Assets, loans and Advances	400
Debentures	450		
Current labilities and Provisions	100		
Total	**1200**	**Total**	**1200**

Mini Case

- FaSon's target capital structure has 50 per cent equity, 10 per cent preference, and 40 per cent debt
- FaSon has Rs. 100 par, 10 per cent coupon, annual payment, noncallable debentures with 8 years to maturity. These debentures are selling currently at Rs. 112
- FaSon has Rs. 100 par, 9 per cent, annual dividend, preference shares with a residual maturity of 5 years. The market price of these preference shares is Rs. 106
- FaSon's equity stock is currently selling at Rs. 80 per share. Its last dividend was Rs. 2.80 and the dividend per share is expected to grow at a rate of 10 per cent in future
- FaSon's equity beta is 1.1, the risk-free rate is 7 per cent, and the market risk premium is estimated to be 7 per cent
- FaSon's tax rate is 30 per cent.
- The new business that FaSon is considering has different financial characteristics than FaSon's existing business. Firms engaged purely in such business have, on average, the following characteristics:
  - Their capital structure has debt and equity in equal proportions
  - Their cost of debt is 11 per cent
  - Their equity beta is 1.5

Mini Case

1. What sources of capital would you consider relevant for calculating the weighted average cost of capital?
2. What is FaSon's post-tax cost of debt?
3. What is FaSon's cost of preference?
4. What is FaSon's estimated cost of equity using the dividend discount model?
5. What is FaSon's estimated cost of equity using the CAPM model?
6. What is FaSon's weighted average cost of capital using CAPM for the cost of equity.
7. What would be your estimate for the cost of capital for the new business?
8. What is the difference between company cost of capital and project cost of capital?

Practice Problems (HW)

1. Azeez Corporations WACC is 12 per cent and its tax rate is 35 per cent. Azeez's pre-tax cost of debt is 14 per cent and its debt-equity ration is 1:1. the risk-free rate is 11 per cent and the market risk premium is 8 per cent. What is the beta of Azeez's equity?
2. The current market price per share of Optimex Ltd. Is Rs. 150. The dividend expected a year from now is Rs. 6 and it is expected to grow at a constant rate of 12 per cent. The floatation cost for the new issue will be 10 per cent. What is the cost of new equity?
3. Acme Ltd.'s equity beta is 1.1. The risk free rate is 8 per cent and the market risk premium is 7 per cent. Acme has a debt equity ratio of 1:2. Its pre-tax cost to debt is 10 per cent and its tax rate is 30 per cent. What is its WACC?

Practice Problems (HW)

4. The costs of specific sources of capital for Elegant Paper Ltd. Are as follows:

Cost of debt = 9 per cent

Cost of preference = 14 per cent

Cost of equity = 17 per cents

The capital structure of Elegant Paper Ltd. Is a follows:

Source of Capital	Book Value (in Million)	Market Value (in Million)
Equity	50	100
Preference	10	9
Debt	80	75
Total	**140**	**184**

Calculate WACC using book value and market value.

THANK YOU!!!